



PROGRAMS FOR DAILY MOBILITY AND FLEXIBILITY

The programs in this chapter can be performed by anyone who is interested in improving flexibility, strength, and strength endurance. In addition, as we age, we all start losing balance. Adding stretching exercises to your daily activities will improve your balance. To make changes to any of these areas, you need to be involved in a regular stretching program, preferably as a daily routine. Improvement will not come in a day or two, but rather after a dedicated effort of several weeks.

You can perform these programs with or without another kind of exercise routine. Stretching on a consistent basis, even without another exercise activity, can bring about changes that not only improve flexibility but also increase strength and strength endurance and improve balance.

As in any other exercise program, progression is an integral part of a successful stretching program. The stretching progression should be gradual, going from a lighter load with less time spent on each stretch to a heavier load with more time spent on each stretch. For these programs, begin with the initial program at the beginner's level and then progress through to the advanced level. You may customize this program according to your current level of experience and flexibility.

Intensity is always a critical factor when you want an exercise program to result in changes and improvements. In a stretching routine, intensity is controlled by the amount of pain associated with the stretch—in other words, how much effort you are putting into stretching the muscles. Using a pain scale from 0 to 10, initial pain is light (scale of 1 to 3) and usually dissipates as the stretching time is extended in each stretching routine. Light stretching occurs when you stretch a particular muscle group only to the point at which you feel the stretch, with an associated light pain. Moder-

ate stretching (scale of 4 to 6) occurs when you start to feel increased, or medium, pain in the muscle you are stretching. In heavy stretching (scale of 7 to 10), you will initially experience a moderate to heavy pain at the start of the stretch, but the pain slowly dissipates as stretching continues.

You will find that heavier stretches rather than lighter stretches provide greater improvements in flexibility and strength. Thus, you are the key to your own success, and how well you are able to monitor stretch intensity and tolerate the pain level determines how quick and significant the improvements will be. Controlling the intensity is the key factor in any exercise program, and this also applies to your stretching program.

Because of the complexity of muscle attachments, many stretching exercises simultaneously affect a variety of muscle groups in the body and stretch the muscle groups around multiple joints. Thus, a small change in body position can change the nature of a stretch on any particular muscle. To get the maximal stretching benefit in any muscle, it is helpful to know the joint movements each muscle can do. Putting the joint through the full range of each motion allows for maximal stretching.

In the programs that appear in this chapter, specific instructions are given relating to the time to hold the stretch and the time to rest between each stretch, as well as the number of repetitions you should do. Follow these instructions in order to get the benefits described. For example, if the instructions indicate you should hold a stretch position for 10 seconds, time or count out the stretch to ensure that you hold it for the recommended time. In addition, you should incorporate only two to four heavier stretching days in each week and have a lighter stretching day between each of the heavier stretching days.

At the beginning of your stretching program, start each stretch with a light stretch and consider this your warm-up. After the initial warm-up stretch, move to your regular program. You need to slowly build up your tolerance to stretching and then move forward in the program as your flexibility improves. Tolerance is built by stretching on a regular basis, the same for any type of exercise program. Stretching is considered a workout just like any other exercise routine.

For a stretch that requires you to sit or lie down, have a soft surface such as a carpet or athletic mat under you. Cushioning makes the exercises more comfortable and enjoyable. However, the cushioning should be firm. A cushion that is too soft will reduce the effectiveness of the stretch.

General Guidelines to Follow for All Stretching Programs

- Include all the major muscle groups of the body in your stretching program.
- Do at least one stretch for each joint movement.
- Before any physical activity, use only light stretches as part of the warm-up.

- After an exercise routine, cool down with light- to medium-intensity stretches.
- If muscles are sore after exercise, perform only light stretches two or three times, with a 5- to 10-second hold and 5 to 10 seconds of rest between each stretch.
- If muscle soreness persists for several days, continue doing light stretches two or three times with a 5- to 10-second hold for each stretch.
- The majority of the stretches should be static in nature.

BASIC STRETCHING PROGRAMS FOR OVERALL DAILY MOBILITY AND FLEXIBILITY

All the stretches in chapters 2 through 8 are best performed in a static manner by holding the stretch for a specified length of time. You get the most benefit when you do these static stretches several times a week at the end of any other workout activity such as jogging or weightlifting. For general overall body flexibility, follow the ability level guidelines as well as the basic guidelines for the stretches detailed in tables 10.1 through 10.4. In tables 10.5, 10.6, and 10.7, you will find stretches for nonsport activities that can be benefited by a stretching program.

Table 10.8 contains a special stretching program shown to help people with either prediabetes or diabetes to lower blood glucose. In 2011, the *Journal of Physiotherapy* published a research study by Nelson, Kokkonen, and Arnall showing that a program of passive static stretches could lower blood glucose by an average of 18 percent after 20 minutes and 26 percent after 40 minutes. These researchers concluded that static stretching is an additional viable activity that can acutely help regulate blood glucose. Moreover, since stretching requires little effort, it appears to be an advantageous treatment for those with reduced physical capabilities. It can also be done without any additional equipment, facilities, or expenses and should easily fit into the repertoire of treatment modalities of any person with diabetes. In addition, since all the stretches in the study were done passively with the help of an assistant, if the person does the stretches actively by himself, the lowering effect on blood glucose should be greater.

Beginner Level

- 1. Hold the stretch position for 5 to 10 seconds.
- 2. Rest for 5 to 10 seconds between each stretch.
- 3. Repeat each stretch two or three times.
- 4. Use an intensity level on the scale from 1 to 3 (light pain).
- 5. Stretch for a total of 15 to 20 minutes each session.
- 6. Stretch two or three times per week.
- 7. Stay on this program at least four weeks before going to the next level.

TABLE 10.1 Beginner Static Stretch Routine

Area	Stretch	Page number
Neck	Neck extensor stretch	176
	Neck flexor stretch	180
Shoulders, back, and chest	Beginner shoulder flexor stretch	146
	Seated shoulder flexor, depressor, and retractor stretch	154
	Beginner shoulder extensor, adductor, and retractor stretch	156
Arms, wrists, and hands	Elbow flexor stretch	112
	Triceps brachii stretch	110
	Forearm pronator stretch with dumbbell	118
	Beginner wrist extensor stretch	122
	Beginner wrist flexor stretch	126
Lower trunk	Supine lower-trunk flexor stretch	90
	Seated lower-trunk extensor stretch	97
	Beginner lower-trunk lateral flexor stretch	100
Hips	Beginner seated hip external rotator stretch	64
	Hip and back extensor stretch	62
	Seated hip adductor and extensor stretch	78
Knees and thighs	Beginner seated knee flexor stretch	40
	Beginner seated knee extensor stretch	50
Feet and calves	Beginner seated toe extensor stretch	12
	Beginner seated toe flexor stretch	16
	Beginner plantar flexor stretch	20

Intermediate Level

1. Hold the stretching position for 15 to 20 seconds.
2. Rest for 15 to 20 seconds between each stretch.
3. Repeat each stretch three or four times.
4. Use an intensity level on the scale from 4 to 6 (moderate pain) two or three times per week.
5. Use an intensity level on the scale from 1 to 3 two or three times per week.
6. Stretch for a total of 30 to 40 minutes each session.
7. Stretch four or five times per week.
8. Stay on this program at least four weeks before going to the next level.

TABLE 10.2 Intermediate Static Stretch Routine

Area	Stretch	Page number
Neck	Neck extensor stretch	176
	Neck flexor stretch	180
Shoulders, back, and chest	Intermediate shoulder flexor stretch	148
	Seated shoulder flexor, depressor, and retractor stretch	154
	Intermediate shoulder extensor, adductor, and retractor stretch	158
	Shoulder adductor, protractor, and elevator stretch	160
Arms, wrists, and hands	Elbow flexor stretch	112
	Triceps brachii stretch	110
	Forearm pronator stretch with dumbbell	118
	Intermediate wrist extensor stretch	124
	Intermediate wrist flexor stretch	128
Lower trunk	Supine lower-trunk flexor stretch	90
	Intermediate lower-trunk lateral flexor stretch	102
Hips	Intermediate seated hip external rotator and extensor stretch	66
	Hip and back extensor stretch	62
	Seated hip adductor and extensor stretch	78
Knees and thighs	Intermediate standing knee flexor stretch	42
	Intermediate lying knee extensor stretch	52
Feet and calves	Beginner seated toe extensor stretch	12
	Beginner seated toe flexor stretch	16
	Beginner plantar flexor stretch	20

Advanced Level

1. Hold the stretch position for 25 to 30 seconds.
2. Rest for 25 to 30 seconds between each stretch.
3. Repeat each stretch five times.
4. Use an intensity level on the scale from 7 to 10 (heavy pain) two or three times per week.
5. Use an intensity level on the scale from 1 to 6 two or three times per week.
6. Stretch for a total of 50 to 60 minutes each session.
7. Stretch four or five times per week.
8. Stay on this level of the program for as long as you want.

TABLE 10.3 Advanced Static Stretch Routine

Area	Stretch	Page number
Neck	Neck extensor stretch	176
	Neck flexor stretch	180
Shoulders, back, and chest	Advanced shoulder flexor stretch	150
	Seated shoulder flexor, depressor, and retractor stretch	154
	Intermediate shoulder extensor, adductor, and retractor stretch	158
	Shoulder adductor, protractor, and elevator stretch	160
Arms, wrists, and hands	Elbow flexor stretch	112
	Triceps brachii stretch	110
	Intermediate wrist extensor stretch	124
Lower trunk	Prone lower-trunk flexor stretch	92
	Advanced standing lower-trunk lateral flexor stretch	104
Hips	Advanced standing hip external rotator stretch	68
	Hip and back extensor stretch	62
	Advanced seated hip adductor stretch	76
Knees and thighs	Advanced seated knee flexor stretch	44
	Advanced kneeling knee extensor stretch	54
Feet and calves	Advanced standing toe extensor stretch	14
	Advanced standing toe flexor stretch	18
	Advanced plantar flexor stretch	22
	Advanced plantar flexor and foot evertor stretch	28

Expert Level

1. Hold the stretch position for 30 to 40 seconds.
2. Rest for 30 to 40 seconds between each stretch.
3. Repeat each stretch five times.
4. Use an intensity level on the scale from 7 to 10 (heavy pain) two or three times per week.
5. Stretch for a total of 50 to 60 minutes each session.
6. Stretch four or five times per week.
7. Stay on this level of the program for as long as you want.

TABLE 10.4 Expert Static Stretch Routine

Area	Stretch	Page number
Neck	Neck extensor stretch	176
	Neck flexor stretch	180
Shoulders, back, and chest	Assisted shoulder and elbow flexor stretch	152
	Assisted shoulder abductor stretch	164
Arms, wrists, and hands	Elbow and wrist flexor stretch	114
	Triceps brachii stretch	110
	Forearm pronator stretch with dumbbell	118
	Intermediate wrist extensor stretch	124
Lower trunk	Prone lower-trunk flexor stretch	92
	Advanced standing lower-trunk lateral flexor stretch	104
Hips	Advanced standing hip external rotator stretch	68
	Advanced seated hip adductor stretch	76
Knees and thighs	Expert raised-leg knee flexor stretch	46
	Advanced supported standing knee extensor stretch	56
Feet and calves	Advanced standing toe extensor stretch	14
	Advanced plantar flexor stretch	22
	Advanced plantar flexor and foot evertor stretch	28

STRETCHING PROGRAMS FOR SPECIFIC DAILY MOBILITY AND FLEXIBILITY CONCERNS

Most of the stretches in the following programs are best performed in a static manner. For these stretches, follow the general recommendations described earlier for your specific level of flexibility. In addition, some of the stretches can be performed in a dynamic manner as a preexercise routine. To perform a stretch dynamically, follow the general guidelines listed in chapter 9 as well as the ability guidelines listed in chapter 11.

Basic Guidelines

1. Hold the stretch position for 30 to 40 seconds for each stretch.
2. Rest for 15 seconds between stretches.
3. Repeat each stretch four times.
4. Use an intensity level on the scale from 1 to 3 (light pain sensation).
5. Do all four stretches on one limb before doing the same stretches on the opposite limb.
6. For the best results, do all of the stretches listed in the tables.
7. You do not need to do the stretches in the order listed. You can do the stretches in an order that best suits your needs.

TABLE 10.5 Stretching Program for Tight or Frozen Shoulder

Area	Stretch	Page number
Shoulders, back, and chest	Dynamic shoulder circumductor stretch	206
	Advanced shoulder flexor stretch	150
	Shoulder adductor stretch	166
	Shoulder internal rotator stretch	168
	Shoulder external rotator stretch	170
	Dynamic shoulder girdle abduction and adduction stretch	204
	Dynamic shoulder flexion and extension stretch	202

TABLE 10.6 Stretching Program for Walking and Standing for Extended Periods

Area	Stretch	Page number
Shoulders, back, and chest	Advanced shoulder flexor stretch	150
	Intermediate shoulder extensor, adductor, and retractor stretch	158
Lower trunk	Standing lower-trunk flexor stretch	94
	Intermediate lower-trunk lateral flexor stretch	102
Hips	Advanced standing hip external rotator stretch	68
	Recumbent hip external rotator and extensor stretch	70
	Hip external rotator and back extensor stretch	72
	Hip and back extensor stretch	62
	Advanced seated hip adductor stretch	76
Knees and thighs	Advanced seated knee flexor stretch	44
	Advanced kneeling knee extensor stretch	54
Feet and calves	Advanced standing toe extensor stretch	14
	Advanced plantar flexor stretch	22
	Advanced plantar flexor and foot evertor stretch	28
	Advanced plantar flexor and foot invertor stretch	34

TABLE 10.7 Stretching Program for Desk and Sitting Jobs or Air and Vehicle Travelers

Area	Stretch	Page number
Neck	Neck extensor stretch	176
	Neck flexor stretch	180
Shoulders, back, and chest	Dynamic shoulder circumductor stretch	206
	Dynamic shoulder flexion and extension stretch	202
	Dynamic shoulder girdle abduction and adduction stretch	204
Hip, lower back	Dynamic hip external and internal rotator stretch	188
	Dynamic hip flexor and extensor stretch	192
	Dynamic hip adductor and abductor stretch	190
Knees and thighs	Dynamic standing knee flexor stretch	194
Trunk	Dynamic trunk lateral flexion	198
	Dynamic trunk rotator stretch	200
Feet and calves	Dynamic plantar flexor stretch	196
	Beginner seated toe extensor stretch	12
	Beginner seated toe flexor stretch	16
	Beginner seated toe extensor and foot evertor stretch	24
	Beginner seated toe extensor and foot invertor stretch	30
	Beginner seated toe flexor and foot evertor stretch	26
	Beginner seated toe flexor and foot invertor stretch	32

TABLE 10.8 Stretches That Lower Blood Glucose, in Order

Area	Stretch	Page number
Knees and thighs	Beginner seated knee flexor stretch	40
Hips	Seated hip adductor and extensor stretch	78
Shoulders, back, and chest	Advanced shoulder flexor stretch	150
Knees and thighs	Intermediate lying knee extensor stretch	52
Hips	Intermediate seated hip external rotator and extensor stretch	66
Shoulders, back, and chest	Intermediate shoulder extensor, adductor, and retractor stretch	158
Knees and thighs	Advanced seated knee flexor stretch	44
Feet and calves	Beginner plantar flexor stretch	20
Shoulders, back, and chest	Shoulder adductor and extensor stretch	162